Li Shen, PhD

Worcester, MA | lshen@clarku.edu | linkedin.com/in/lily-shen

RESEARCH INTERESTS

International trade; quantitative spatial economics; market integration; regional development; urban and population geography; industrial policy; China and emerging Asia.

EDUCATION

Clark University — **Worcester, MA**
Ph.D. in Economics — *2026*

The State University of New York at Buffalo — **Buffalo, NY**
M.A. in Economics — *May 2019*

Valparaiso University — **Valparaiso, IN**
M.S. in International Economics and Finance — *May 2018*

Chongqing University — **Chongqing, China**
B.Mgmt. in Accounting — *June 2014*

WORKING PAPERS

The Local Multiplier Effects of Trade: A Quantitative Analysis of China — Job Market Paper
with Kensuke Suzuki, Xiaocong Xu, and Junfu Zhang

- Develops a multi-sector quantitative spatial model with inter-provincial input-output linkages, non-tradable production, and labor mobility, calibrated to China's 2017 MRIO table; parameters recovered via structural inversion from MRIO, ICIO, and Customs micro-data.
- Establishes that local trade multipliers are equilibrium objects shaped by the structure of domestic production linkages rather than universal parameters, and identifies the own-trade share—not raw export exposure—as the structural organizing statistic for multiplier heterogeneity.
- Extends the quantitative spatial economics framework, typically used for national welfare analysis, to the mechanics of sub-national shock amplification; counterfactual simulations yield model-implied local GDP multipliers ranging from 0.30 to 1.02 across provinces, with inland inward-oriented economies amplifying external shocks through non-tradable service demand.

High-Speed Railway and Market Integration: Evidence from Vegetable Price Dispersion in China — Working Paper

- Constructs the largest daily-frequency panel of Chinese vegetable wholesale prices to date: 3.99 million city-pair-month observations covering ten commodities across 117 prefecture-level cities (2014–2022), merged with AutoNavi smartphone-based intercity passenger mobility indices.
- Leverages two institutional features—China's Green Passage highway-toll exemption (which fixes variable freight costs for fresh vegetables) and prohibitive HSR freight tariffs for bulk produce—to study a setting in which differences in HSR connectivity reflect variation in passenger rather than freight costs.
- Documents that HSR-connected city pairs exhibit 2–5% lower intercity price dispersion; passenger-mobility controls absorb the direct HSR coefficient, and cross-product heterogeneity (larger effects for bulky, low-value commodities) argues against a freight substitution channel, pointing to passenger mobility and trader interaction as the operative mechanism.

METHODS AND DATA EXPERTISE

- **Methods:** causal inference; shift-share IV; panel-data econometrics; event studies; staggered-adoption designs; gravity models; quantitative spatial modeling; input-output analysis; counterfactual simulation.
- **Data:** Chinese MRIO and ICIO tables; Chinese Customs data; AutoNavi mobility data; high-frequency wholesale price data; geospatial population rasters; GIS-based regional datasets.
- **Programming:** Python; Stata; R; MATLAB; SQL; SAS; GeoPandas; ArcGIS; Tableau; LaTeX.
- **Languages:** Mandarin Chinese (native); English (fluent).

APPLIED PROJECTS

Pricing and Promotion Decision Engine — Demand Estimation, Optimization, and Deployment — 2025–Present
Live app: pricing-promotion-decision-engine.streamlit.app *Code:* github.com/Li-Shen-Clark/pricing-promotion-decision-engine

- Built an end-to-end pricing decision-support application on 6.6M rows of Dominick's Finer Foods scanner data; cleaned to a 4.65M-row panel covering 486 UPCs, 93 stores, and 366 weeks.
- Estimated a log-log fixed-effects demand model with Duan smearing retransformation to forecast demand responses to price and promotion changes.

- Implemented counterfactual price and promotion simulations plus constrained profit optimization under cost, inventory, and margin constraints.
- Designed A/B validation plans with power analysis for raise-and-test pricing candidates; deployed as a Streamlit app with modular Python code, reproducible requirements, GitHub auto-deployment, and pytest tests.

China Population Density Atlas — Interactive Geospatial Visualization 2022–Present

Live app: pop-density-spatial.streamlit.app

- Processed 19 years of 5 km-resolution population-density rasters (2002–2020) covering the full territory of China into a harmonized annual panel.
- Underlying analysis documents increasing spatial agglomeration and expansion of dense urban corridors, with little evidence of province-level convergence.
- Deployed an interactive Streamlit application in 2025 for map-based exploration of density, agglomeration, and urban-corridor metrics at the grid-cell and province levels.

MACHINE LEARNING COMPETITIONS

Home Credit — Credit Risk Model Stability (Kaggle Silver Medal) Feb–May 2024

Stack: Python, Polars, scikit-learn, LightGBM, CatBoost

- Co-developed a Polars lazy-execution ETL pipeline for multi-depth relational data; designed aggregation features (max, last, mean) capturing borrower behavioral trends; optimized directly for the headline Gini Stability metric to prevent performance drift over time.
- Implemented correlation-based feature grouping (Pearson > 0.8) for multicollinearity control and type-downcasting memory optimization (>50% reduction).

ICR — Identifying Age-Related Conditions (Kaggle Bronze Medal) May–Aug 2023

Stack: Python, TabPFN, XGBoost, imbalanced-learn

- Designed a stratified resampling strategy using auxiliary Greek metadata (RandomOverSampler on Alpha-defined latent subgroups) to preserve population structure under severe class imbalance.
- Co-developed a voting ensemble combining parametric (XGBoost) and non-parametric (TabPFN) models under a custom Balanced Log Loss metric; calibrated decision thresholds to minimize false negatives.

TEACHING EXPERIENCE

Clark University School of Management **Worcester, MA**

Graduate Teaching Assistant *Fall 2020–Spring 2023*

- **Investments (FIN 5401):** Instructed graduate students on equity valuation, DCF modeling, and portfolio risk-return analysis.
- **Computational Finance (FIN 5216):** Led graduate-level computational sessions on Monte Carlo methods, stochastic processes, and Black-Scholes pricing using R and Excel.
- **Case Studies in Derivatives (FIN 5310):** Guided students in quantifying financial risk, analyzing gold price risk management, and evaluating FX hedging strategies for global operations.
- **Investment Strategy (FIN 5203):** Supported instruction on advanced portfolio management and asset allocation strategies.

Clark University Department of Economics **Worcester, MA**

Teaching Assistant *Fall 2023–Spring 2025*

- **Macroeconomic Theory (ECON 206):** Led undergraduate problem-solving sessions on the Solow growth model, the IS-LM and AS-AD frameworks, Phillips-curve dynamics, monetary and fiscal policy transmission, and open-economy macroeconomics.
- **Economics and the World Economy (ECON 10):** Supported undergraduate instruction on comparative advantage, exchange-rate regimes, international capital flows, economic development, and trade-policy debates.

HONORS AND AWARDS

Sheftel Award for Research Excellence, Clark University **2025**

Veendorp Best Field Paper Award, Clark University **2022**

Doctoral Fellowship, Clark University **2019–2026**

Distinguished Student Award (Graduate School), Valparaiso University Alumni Association **2018**

REFERENCES

Dr. Junfu Zhang
Professor of Economics
Clark University
juzhang@clarku.edu

Dr. Kensuke Suzuki
Assistant Professor of Economics
Clark University
KSuzuki@clarku.edu

Dr. Moshi Alam
Assistant Professor of Economics
Clark University
mdalam@clarku.edu